

Exp B: sliding $\hat{\beta}_t$ on alternating-genre text

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1 Question

The project proposal’s stretch goal was: do per-context sliding $\hat{\beta}_t$ estimates beat batch fitting on heterogeneous text? After Exp A, the calibration framing of this question is moot — the filter is not a Guo-style temperature scaler (cf. Exp A findings). The remaining testable question: *does the sliding $\hat{\beta}_t$ signal track corpus genre transitions at all?*

2 Setup

Construct one mixed sequence by interleaving 4 WikiText-103 articles and 4 WritingPrompts stories, each truncated to 256 tokens: [wiki, WP, wiki, WP, wiki, WP, wiki, WP] $\rightarrow$ 2048 tokens total. Run the streaming Laplace filter with $\sigma_0 = 0.5$ and memory_decay $\gamma \in \{0.90, 0.95, 0.98, 0.99, 1.00\}$. Within each chunk, exclude a 20-token boundary window on each side before averaging (per the seam-confound caveat raised in the Exp A audit).

3 Result

Table 1: Mean $\log \hat{\beta}$ per genre over boundary-stripped chunk interiors, and mean pre-decoding model entropy H per genre.

γ	mean $\log \hat{\beta}$ wiki	mean $\log \hat{\beta}$ WP	WT – WP	mean H wiki	mean H WP
0.90	−0.084	−0.138	+0.054	1.686	3.024
0.95	−0.092	−0.142	+0.051	1.686	3.024
0.98	−0.115	−0.143	+0.028	1.686	3.024
0.99	−0.142	−0.140	−0.002	1.686	3.024
1.00	−0.458	−0.463	+0.005	1.686	3.024

Three observations.

(a) The damped filter does see a small genre signal. At $\gamma = 0.90$ – 0.95 , the within-chunk mean of $\log \hat{\beta}$ is ≈ 0.05 log units higher on WikiText than on WritingPrompts. This is consistent in sign with the earlier natural.text/findings result that the filter’s $\hat{\beta}$ runs slightly negative on WritingPrompts and near zero on WikiText. The signal weakens as $\gamma \rightarrow 1$.

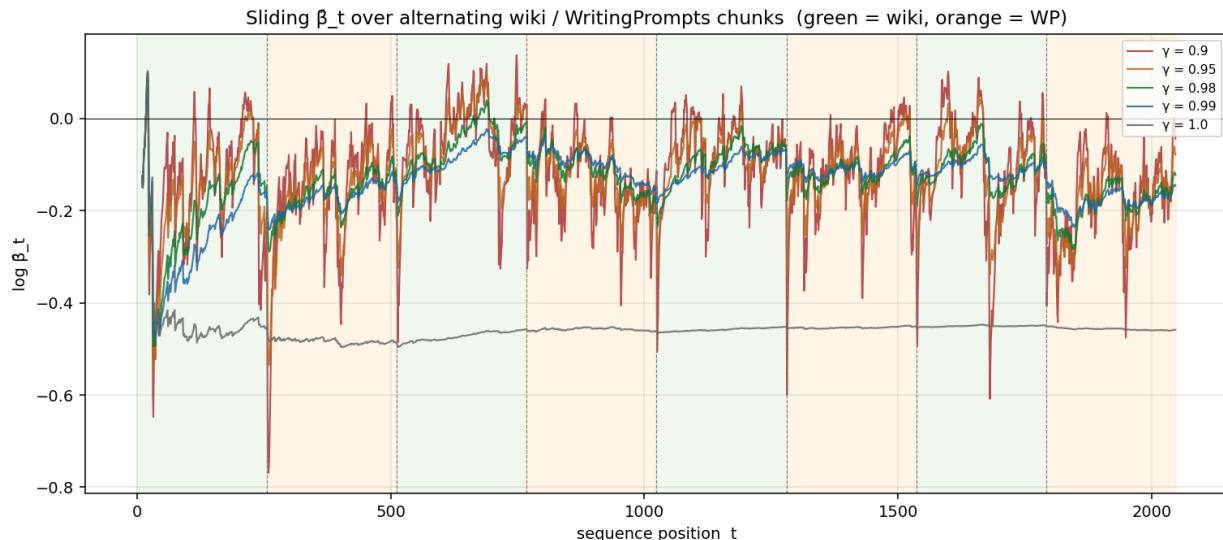


Figure 1: $\log \hat{\beta}_t$ as a function of sequence position t for each γ . Background shading is genre: green = WikiText, orange = WritingPrompts. $\gamma = 1.00$ (no decay, grey) just drifts to ≈ -0.45 and stops tracking content — the unconditional pooling bias from Exp A. Damped filters oscillate.

(b) The genre signal is dwarfed by the entropy signal. Pre-decoding model entropy H differs by ≈ 1.3 nats between the two genres (1.69 on wiki, 3.02 on WP). That gap dwarfs the 0.05-log-unit filter gap. Whatever the filter is doing here is mostly tracking some moderate-bandwidth signal about token predictability, not a sharply genre-distinct quantity.

(c) $\gamma = 1.00$ saturates to the pooled-streaming bias. With no decay, the filter accumulates score and Fisher across the full 2048-token sequence and lands at $\log \hat{\beta} \approx -0.46$ for every chunk — the same one-Newton-step-from-prior-mode artifact established in Exp A. Memory decay is necessary if one wants the filter to retain *any* per-context sensitivity.

4 Bottom line

The sliding $\hat{\beta}_t$ filter at $\gamma = 0.90$ – 0.95 detects a small (≈ 0.05 -log-unit) signal that is consistent in sign with WritingPrompts being apparently lower- $\hat{\beta}$ than WikiText. This is not strong enough to support a per-context calibration claim on top of Exp A’s finding that the filter is not a Guo-style calibrator at all. At $\gamma = 1.00$ the filter saturates to its accumulated-bias value and loses all sensitivity.

What it suggests: the filter’s signal here is plausibly content-driven (higher prefix entropy $\rightarrow$ score and Fisher each carry the corresponding content signature) rather than genuinely about the model’s local belief about its decoder. Disentangling this from a true decoder-belief signal would require a controlled experiment varying β_{dec} within a single genre while holding content fixed — which Exp C attempted and found dominated by garbage-in/garbage-out off-distribution drift rather than decoder belief.

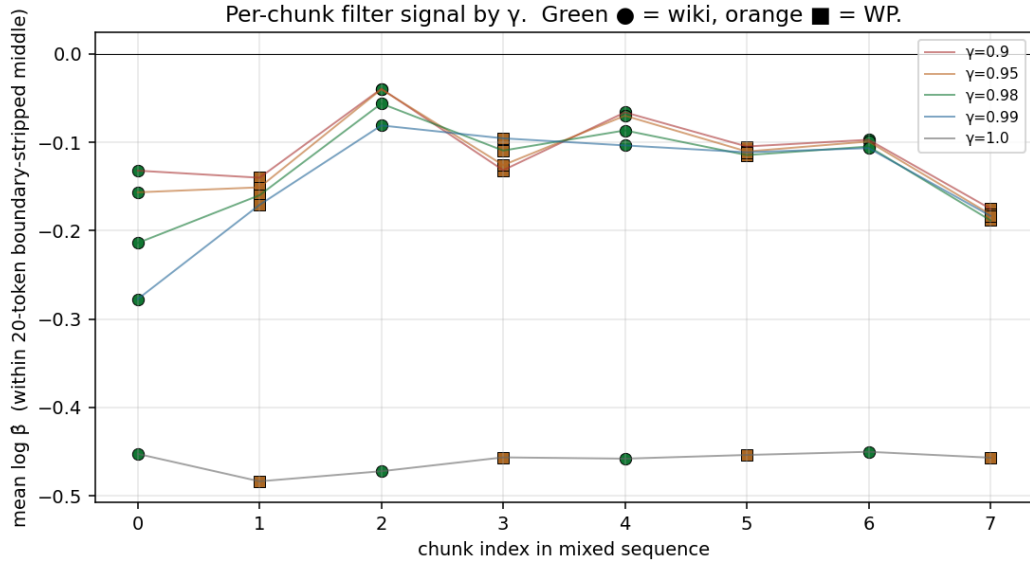


Figure 2: Per-chunk mean $\log \hat{\beta}$ across the 8 chunks in order. $\gamma = 1.00$ ignores the genre structure entirely (it has saturated to the unconditional pooled-streaming bias). Damped filters show some chunk-to-chunk variation, but the WT–WP gap is small (≈ 0.05 log units), comparable to within-chunk noise.